

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5.	(a)			ATP and NADPH are produced in the light-dependent reactions (1) Any one (x1) from {Electrons/H atoms/reduced coenzymes/ NADPH₍₂₎} are needed for GP to TP (1) - ATP needed to {re-phosphorylate/ provide energy/ regenerate RuBP/ for GP to TP} (1) - So that the amounts of ATP and NADPH are not rate limiting in light-independent reactions (1)	2			2		
	(b)	(i)		B is correct (1) Any one (x1) from Because {glycerate phosphate/GP} appears first (1) Reference to data e.g. GP is present at 5secs (1)			2	2		2
		(ii)		- Without nitrate X is not {produced/ present} / Compound X needs nitrate to be formed/ X must contain nitrogen (1) - X is amino acid/ nucleotide/ organic base / protein/ enzyme/ ATP/ named examples of above (1) Reject nitrate		1	1	2		
	(c)	(i)		<u>Stroma</u> (in chloroplast) (1)	1			1		
		(ii)		Accept values 1.3 to 1.75µm = 2 marks Award 1 mark for $\frac{23 \text{ or } 24}{7,8 \text{ or } 9} \times 500$ Values in nm between 1300 and 1750		2		2	2	
	(d)	(i)		Increased yield/ faster growth/ {bigger/ sweeter} crops/ increased photosynthesis (1) Because plants will take in more CO ₂ / more CO ₂ is available to the plant (1)		1	1	2		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		- Increased photosynthesis / decreased atmospheric CO₂/ more CO₂ absorbed (1) - Will improve prospects for endangered species (1) (Because less CO₂ results in) less greenhouse effect / less climate change/ global warming/ higher yield leading to less habitat destruction (1)		3		3		1
				Question 5 total	3	7	4	14	2	5